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Hematological alterations in *Clarias batrachus* (L.) exposed to sublethal concentrations of commercial chlorpyrifos

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Abstract- Pesticide pollution is a pervasive global issue, and the consumption of pesticides is growing daily due to the development of high-yield crop varieties. Pesticides used in agricultural areas adversely affect fish and other aquatic organisms. Barely 0.1 percent of pesticides successfully reach their intended target species, while the majority is disseminated into the surrounding ecosystem. Surface runoff of pesticides into aquatic ecosystems can be severely toxic to aquatic organisms. Though banned by the USEPA in 2021, chlorpyrifos, an organophosphorus pesticide is widely used in the Indian market. The breakdown and metabolism of Chlorpyrifos leads to the production of metabolites that may exhibit various toxicity at the biogenic level. The present study seeks to asses hematological alterations induced by sublethal exposure of commercial formulation of chlorpyrifos (CPF) in native catfish, *Clarias batrachus*. This was achieved by elucidating blood biomarkers in a time and dose dependent manner. Treatment of the fish with chlorpyrifos at two sub-lethal doses compared to the control group led to various changes in the hematopoietic parameters in the fish. Among the notable abnormalities were significant decrease in parameters such as hemoglobin (g/dL), red blood cell count (106/mm^3), hematocrit (PCV) (%), mean corpuscular hemoglobin concentration (g/dL). However, the mean corpuscular volume (fL), mean corpuscular hemoglobin (pg) and blood glucose (g/dL) demonstrated an opposite order. These disruptions reflect clear physiological stress. Findings underscore CPF's capacity to disturb blood parameters in this species and suggest potential interventions.

Keywords: Hematocrit, blood glucose, organophosphate, *Clarias batrachus*, macrocytic anemia

INTRODUCTION

Intensified agricultural activities have heightened agrochemical contamination in aquatic ecosystems. Pesticides reach freshwater systems primarily via surface runoff, leaching, and unintended spills, creating substantial risks for non-target aquatic life including fish.[1] Chlorpyrifos, a broad-spectrum organophosphate insecticide, is extensively applied in agriculture to manage diverse pests. However, its high toxicity to aquatic life has led to frequent detections in surface waters adjacent to farmlands.[2]

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Fish serve as excellent bioindicators of aquatic pollution due to their high sensitivity to water quality fluctuations and their ability to bioaccumulate contaminants from surrounding environments.[3] *Clarias batrachus* (L.), known as the walking catfish, is a valuable freshwater teleost with substantial commercial and nutritional importance across India and Southeast Asia. Its robust physiology and ecological significance position it as an excellent model species for toxicological research. However, sublethal pesticide exposures trigger physiological stress responses and interfere with normal metabolic functions in organisms without causing immediate mortality.[4]

Hematological parameters serve as highly reliable biomarkers of pesticide-induced stress in fish, as blood plays essential roles in respiration, nutrient delivery, immunity, and waste elimination. Alterations in hemoglobin levels, red blood cell counts, white blood cell numbers, and hematocrit values directly indicate pollutant effects on hematopoietic tissues and immune responses.[1] Organophosphate pesticides like chlorpyrifos primarily exert their toxicity through acetylcholinesterase inhibition, while also causing secondary effects such as oxidative stress, anemia, and immunosuppression.[5]

Therefore, in the present study assessment of hematological alterations in *Clarias batrachus* exposed to sublethal concentrations of commercial chlorpyrifos is done. The study aims to provide valuable information on the sublethal toxic effects of this pesticide essential for environmental monitoring, ecological risk evaluation, and development of pesticide regulations to safeguard aquatic biodiversity and human health.

MATERIALS & METHODS

Chemical toxicant

In the current investigation, a commercial chlorpyrifos formulation (20% EC), marketed as Eldrin^{TC} and produced by Crystal Crop Protection Ltd., was sourced from the local agrochemical market in Ranchi, Jharkhand. These preparations were deliberately chosen to mirror formulations routinely employed by regional farmers, ensuring ecological relevance, and were employed in experimental procedures without any alterations.

Animal Collection and Maintenance

Fish, *Clarias batrachus,* of uniform size (length: 17.1 ± 1.4 cm; weight: 35.7 ± 7.7 g) were obtained from a local supplier in Ranchi, Jharkhand. The study was carried out in the wet laboratory of the University Department of Zoology, Ranchi University, Ranchi, Jharkhand. Initially, the fish received prophylactic treatment via two successive 2-min dips in 0.05% potassium permanganate ($KMnO_{,,}$) solution to prevent dermal infections. Subsequently, they were acclimatized for 15 days in 20-L glass aquaria containing dechlorinated tap water prior to experimentation. The experiment was conducted under a natural photoperiod (13 h light: 11 h dark). Water quality parameters, measured as per APHA (2005) guidelines, included temperature (24.93 ± 1.8 °C), pH (7.07 ± 0.33), dissolved oxygen (8.08 ± 1.6 mg/L), total dissolved solids (240 ± 7.54 ppm), and alkalinity (131.39 ± 4.09 mg/L). The fish were fed twice daily with a commercial diet and were subjected to a 24-hour fasting period prior to the initiation of the experiment.

Experimental design

The 96-hour LC_{50} value for commercial-grade chlorpyrifos in *Clarias batrachus* was determined to be 1.29 mg/L.[6] The fishes were divided into three experimental groups, with each group containing three specimens and three replicates were established for each group. *Group I* served as control. Fishes were exposed to uncontaminated toxicant free tap water. *Group II* included fishes exposed to sublethal concentration of chlorpyrifos at a dose 0.129 mg/L (1/10 of LC_{50}-96 h). *Group III* included fishes exposed to sublethal concentration of chlorpyrifos at a dose 0.258 mg/L (1/5 of LC_{50}-96 h). Hematological analyses of *Clarias batrachus* were conducted following sublethal chlorpyrifos exposure at intervals of 7, 14, 21, and 28 days. Assessments were carried out using three fish per group at each time point.

Collection of Blood

Fishes were randomly selected from control and experimental tanks for blood collection. At first, fishes were anesthetized using ether. Then, blood samples were collected from the caudal vein or directly puncturing the heart. Blood (1 mL) was transferred to a microtube containing aqueous solution of EDTA as anticoagulant and immediately used for hematological examination.

Hematological Analysis

Hematological parameters included the analysis of hemoglobin (Hb), red blood cell (RBC) count, packed cell volume (PCV), mean corpuscular volume (MCV), mean corpuscular hemoglobin (MCH), mean corpuscular hemoglobin concentration (MCHC) for control and chlorpyrifos treated fishes using automated hematology analyzer. Glucose was estimated using glucometer.

Statistical Analysis

Statistical analysis was performed using Student's t-test to evaluate differences between control and chlorpyrifos-treated groups. Significance was determined at $p < 0.05$. Data are expressed as mean ± standard deviation (SD).

RESULTS

Fish hematology serves as a valuable indicator of overall health status. Various blood parameters—including total red blood cell (RBC) count, hemoglobin (Hb) content, packed cell volume (PCV), mean corpuscular volume (MCV), mean corpuscular hemoglobin (MCH), and mean

corpuscular hemoglobin concentration (MCHC)-have been widely employed as biomarkers of pesticide contamination in aquatic ecosystems.

This study demonstrated pronounced hematological disruptions in the blood profile of *Clarias batrachus* subjected to chronic sublethal chlorpyrifos exposure. Fish maintained in clean, uncontaminated water (control) exhibited no statistically significant fluctuations across the measured parameters, confirming baseline stability under laboratory conditions.

In sharp contrast, chlorpyrifos-treated groups displayed consistent and dose-dependent alterations. Notably, hemoglobin (Hb) concentration, red blood cell (RBC) count and packed cell volume (PCV) exhibited significant declines ($p < 0.05$) relative to controls at all exposure durations (*Figure 1*). Similarly, mean corpuscular hemoglobin concentration (MCHC) showed a marked reduction ($p < 0.05$) (*Figure 1*). Conversely, mean corpuscular volume (MCV) and mean corpuscular hemoglobin (MCH) values rose significantly ($p < 0.05$) in exposed fish (*Figure 1*). Plasma glucose levels also surged significantly ($p < 0.05$) (*Figure 2*). These patterns underscore chlorpyrifos's disruptive impact on fish erythropoiesis and metabolic homeostasis.

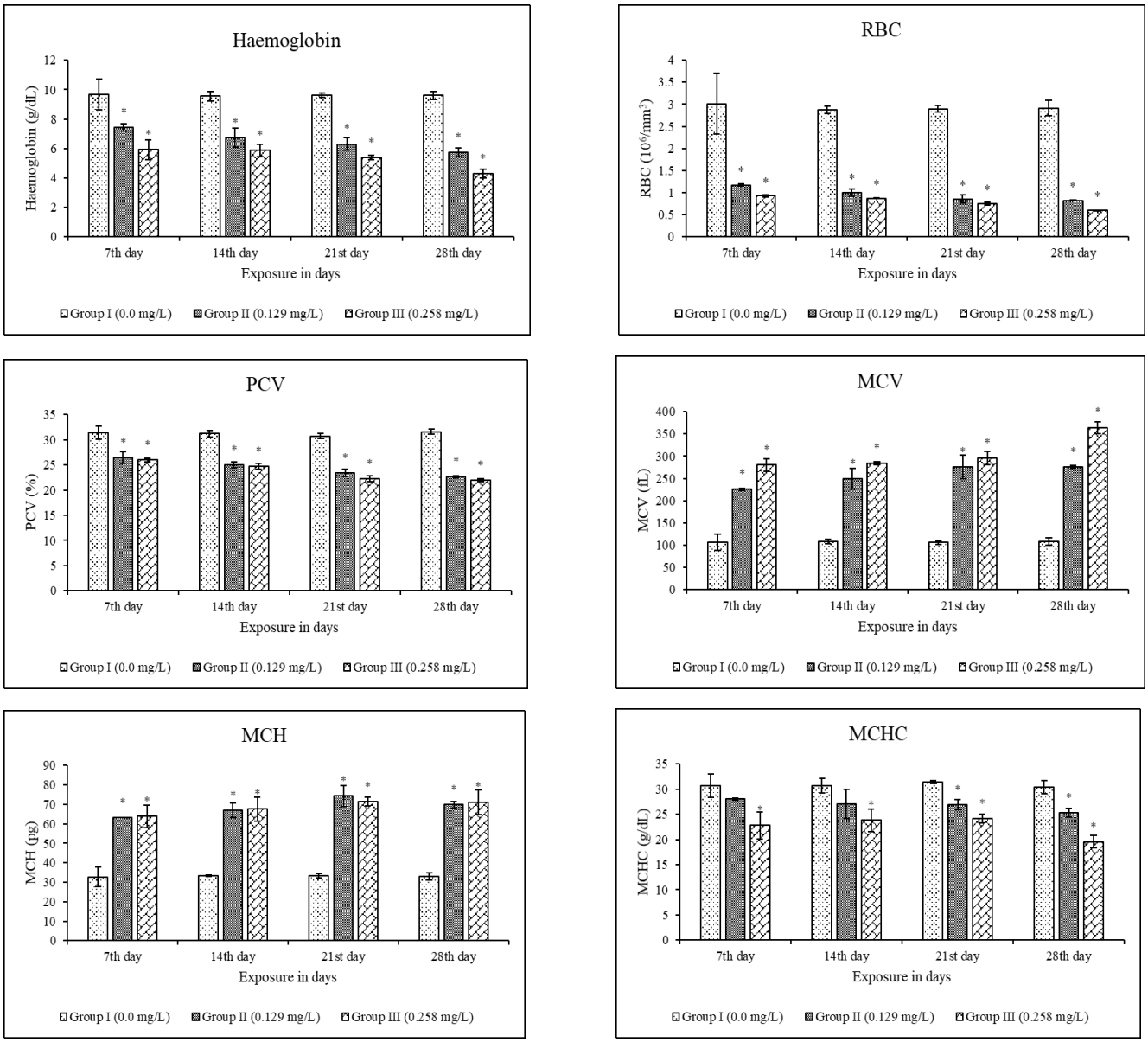


Figure 1: Sublethal effects of chlorpyrifos exposure on hemoglobin, RBC count, PCV, and derived red cell indices (MCV, MCH, MCHC) *in Clarias batrachus*.

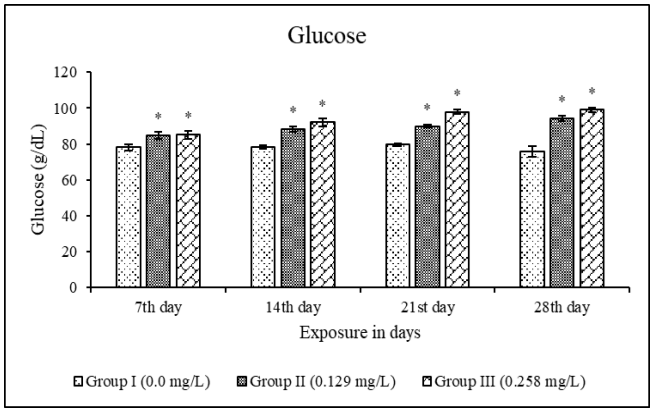


Figure 2: Sublethal effects of chlorpyrifos exposure on blood glucose *in Clarias batrachus*.

DISCUSSION

In the present study, *Clarias batrachus* exposed to sublethal doses of chlorpyrifos for 28 days exhibited pronounced changes in hematological parameters (*Figure 1, 2*). Hematological parameters serve as key indicators of health status in many fish species and are widely used to assess stress conditions such as disease, hypoxia, and exposure to metals and other pollutants.[7] In the exposed fish, blood parameters such as RBC, Hb, PCV, and MCHC exhibited a decreasing trend, whereas MCH, MCV, and glucose levels showed an increasing trend following chlorpyrifos exposure.

In our finding, a time dependent significant decline*(*$p<0.5$*)* in RBCs count, Hb and PCV was observed among both the treatment groups compared to control group, which are in accordance with the study of Hossain *et al.,* (2022)[8] who also testified significant reduction in RBC count, Hb level and PCV in *Oreochromis niloticus* treated with chlorpyrifos during chronic toxicity. RBCs along with Hb play a key part in providing oxygen to various parts of the body, while packed cell volume (PCV) is the percentage of RBCs in the blood which depend upon the number as well as size of RBCs. Therefore, any change in the size and number of RBCs or Hb change the PCV percentage. *Clarias batrachus* exposed to chlorpyrifos showed a significant reduction in RBC and Hb levels resulting in anemia. This condition likely stems from inhibited erythropoiesis and hemosynthesis, coupled with accelerated erythrocyte destruction in hemopoietic organs. Our results are in good concurrence with the earlier works of Ghayyur *et al.,* (2021)[9] on *Cirrhinus mrigala* after sublethal exposure to chlorfenapyr, acetamiprid and dimethoate and Velmurugan *et al*., (2019)[10] on *Anabas testudineus* after exposure to chlorpyrifos.

Blood cell indices such as MCV, MCH, and MCHC can disrupt fish homeostatic balance in a reversible fashion and show high sensitivity to targeted environmental pollutants.[11] Erythrocyte indices such as MCV, MCH, and MCHC provide insights into the size, shape, and hemoglobin content of red blood cells. These parameters help classify anemias as normocytic, macrocytic, or microcytic based on morphological changes.[12] This study revealed significant elevations ($p<005$) in MCV and MCH, coupled with a decline in MCHC, in fish exposed to chlorpyrifos compared to controls over the entire experimental period. The MCHC reduction lacked statistical significance up to day 7 but became significant from day 14 onwards. The observed hematological changes in this study likely represent a compensatory defense against chlorpyrifos toxicity, which reduces RBC, Hb, and PCV levels while disrupting metabolic and hemopoietic functions in fish, resulting in elevated MCV and MCH. These elevations suggest macrocytic anemia. Such shifts reflect impaired erythropoiesis and cell swelling under pesticide stress, consistent with pollutant-induced erythrocyte alterations in species like *Clarias batrachus*.[13] The present work is in agreement with the work of Latha and Mohan, (2018)[14] following exposure to *Glossogobius giuris* to 0.25 and 0.5 ppm of Malathion, and attributed these changes to the swelling of RBCs. Complying to the view presented above a rise in the MCV and MCH level in *Oreochromis niloticus* treated with chlorpyrifos was reported by Hossain *et al.,* (2022)[8].

The present study found decreased MCHC values in fish after chlorpyrifos exposure. Chauhan *et al.,* (2025)[15] similarly documented that organophosphate pesticide profenofos induce significant MCV increase coupled with MCHC reductions in *Labeo rohita*. The observed MCHC decline confirms lower hemoglobin concentrations within erythrocytes. Banaee, (2013)[1] similarly documented that organophosphorus compounds induce significant increment of MCV associated with a decrease in MCHC.

In the present study, glucose levels increased significantly ($p<0.05$) in *Clarias batrachus* following chlorpyrifos exposure. This hyperglycemia likely stems from accelerated glucose catabolism to fulfill heightened energy demands triggered by pesticide-induced stress, including enhanced gill ventilation, osmoregulatory strain,

and activation of the hypothalamic-pituitary-interrenal (HPI) axis, which elevates cortisol and mobilizes hepatic glycogenolysis.[9] Such responses represent classic secondary stress indicators in fish toxicology. Chlorpyrifos inhibits acetylcholinesterase, disrupting neurotransmitter balance and prompting catecholamine surges (e.g., epinephrine) that further stimulate glycogen breakdown and gluconeogenesis. Similar patterns occur across species like tilapia and rohu under organophosphate exposure.[13,16] Chhaba *et al*., (2024)[16] also reported an increase in glucose level in *Pangasianodon hypophthalmus* exposed to Chlorpyrifos (20% EC). Elevated plasma glucose level was also reported by Mukherjee *et al*., (2022)[17] in *Clarias batrachus* after 2, 4, 6-Trichlorophenol exposure.

CONCLUSION

In the present study, sublethal exposure of commercial chlorpyrifos in *Clarias batrachus* triggered pronounced hematological disruptions, characterized by decreased RBC counts, hemoglobin, PCV, and MCHC values coupled with elevated MCV, MCH, and serum glucose concentrations. These time- and dose-dependent shifts indicate macrocytic anemia resulting from inhibited erythropoiesis, accelerated erythrocyte destruction, and stress hyperglycemia via HPI axis stimulation. These results in impairments in oxygen delivery, immune competence, and metabolic balance. *C. batrachus* proves an excellent bioindicator species for organophosphate monitoring in Indian freshwater habitats. Urgent regulatory action targeting agricultural runoff reduction remains essential to safeguard catfish populations and aquatic ecosystem health.

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